

## IMPLEMENTATION DETAILS & SNIPPETS

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

### 1. Backend Routing Code (Flask)

The controller logic utilizes Flask routes to accept POST data, process it, and render the results template. Below is an excerpt of the prediction pipeline:

- Extract soil parameters from the HTML form request.
- Transform the input list using the StandardScaler pipeline.
- Perform inference using the Random Forest classifier.
- Predict estimated yield and categorise soil with the KMeans model.
- Append details to datasets/user\_predictions.csv and render results.

### 2. Model Training Script (train\_models.py)

The training pipeline loads datasets, splits them (80/20 train/test split), fits the StandardScaler, trains the classifiers, outputs charts, and serializes files using pickle.